

Trading Challenge

Quantitative Strategy Research Report

Challenge Overview

This document presents our systematic research and development process for constructing, evaluating, and combining quantitative trading strategies across a universe of 20 equities. Rather than treating the tasks as isolated exercises, we approached them as components of a broader portfolio construction framework focused on signal generation, ranking, allocation, and adaptive strategy selection.

1 Project Description

We were provided with OHLC market data for 20 stocks and designed a collection of trading signals that transform historical price information into daily portfolio weights. The first 3500 business days were utilized for model development and calibration, while the following 500 days were reserved for validation and robustness testing.

To ensure realistic evaluation, all portfolio decisions were generated exclusively using information available up to the previous trading day, thereby eliminating look-ahead bias and preserving the integrity of out-of-sample testing.

2 Task 1: Individual Alpha Strategies

2.1 Strategy 1: Weekly Return Reversion Model

We computed weekly returns over the previous 50 completed trading weeks and estimated the mean historical return for every stock. Securities were ranked according to their average weekly performance.

- Top 6 performers received equal short allocations.
- Bottom 6 performers received equal long allocations.
- Remaining securities received zero weight.

The hypothesis behind this model was that extreme historical winners may exhibit mean-reversion characteristics while laggards may experience recovery.

2.2 Strategy 2: Moving Average Divergence

We compared short-term momentum against longer-term price trends using:

- Long-Term Moving Average (30 days)
- Short-Term Moving Average (5 days)

Relative divergence was computed as:

$$\frac{SMA - LMA}{LMA}$$

Stocks exhibiting the strongest positive divergence were shorted, while those showing the largest negative divergence were purchased.

2.3 Strategy 3: Rate of Change Momentum

A seven-day Rate of Change (ROC) indicator was calculated:

$$ROC = 100 \times \frac{P_t - P_{t-7}}{P_{t-7}}$$

The strategy identified securities that had reached extreme short-term momentum conditions and allocated capital toward anticipated reversals.

2.4 Strategy 4: Support and Resistance Framework

We estimated dynamic support and resistance bands using a 21-day moving average and volatility envelope:

$$Resistance = SMA_{21} + 3\sigma$$

$$Support = SMA_{21} - 3\sigma$$

Portfolio construction favored securities trading unusually close to support while simultaneously shorting securities approaching resistance.

2.5 Strategy 5: Stochastic Oscillator Allocation

The stochastic indicator was used to evaluate relative positioning within the recent trading range:

$$\%K = 100 \times \frac{Close - Low_{14}}{High_{14} - Low_{14}}$$

The most overextended securities received short allocations while the most oversold securities received long allocations.

3 Task 2: Adaptive Ensemble Framework

After constructing the five base strategies, we developed an ensemble framework that dynamically selected a single strategy for deployment on any given trading day.

Rather than averaging signals, we evaluated the historical behavior of each strategy and determined which methodology demonstrated the strongest recent effectiveness. This adaptive selection process enabled the portfolio to respond to changing market regimes while maintaining strict adherence to information availability constraints.

Key objectives included:

- Maximizing out-of-sample performance.
- Preserving robustness across market environments.
- Preventing overfitting through validation procedures.
- Maintaining fully automated decision generation.

4 Task 3: Transaction-Cost Aware Optimization

To better approximate real-world implementation, we extended the ensemble system to incorporate transaction costs.

Turnover was defined as:

$$Turnover = \sum |w_t - w_{t-1}|$$

Net performance was evaluated using:

$$Net\ Returns = Returns - 0.01 \times Turnover$$

This extension encouraged smoother transitions between strategies and penalized unnecessary portfolio rebalancing. The resulting framework balanced signal quality against implementation efficiency.

5 Research Methodology

Our development process followed the following sequence:

1. Feature engineering from historical price data.
2. Construction of independent trading signals.
3. Portfolio weight generation.
4. Backtesting and validation.
5. Ensemble strategy development.
6. Transaction-cost optimization.
7. Out-of-sample robustness analysis.

Throughout the project, we prioritized reproducibility, statistical discipline, and realistic market assumptions.

6 Performance Evaluation

The final evaluation framework focused on:

- Risk-adjusted returns.
- Portfolio stability.
- Transaction efficiency.
- Out-of-sample robustness.
- Implementation quality.
- Innovation in signal design and ensemble construction.

7 Conclusion

This project evolved from a collection of standalone trading rules into a comprehensive quantitative research pipeline. By combining multiple alpha-generation methodologies with adaptive strategy selection and transaction-aware optimization, we developed a framework capable of operating under realistic market constraints while maintaining a systematic and data-driven investment process.